PRATHYUSHA ENGINEERING COLLEGE

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PROJECT REPORT

TITLE: Multi-Channel Industrial Sensor Data Selector

SUBMITTED BY

S.No	Name	Register Number
1	Abishal	111425104002
2	Abishek	111425104003
3	Rithesh	111425104008
4	Govardhan	111425104006

SEMESTER : III

YEAR : II/A

COURSE CODE / NAME : 241GES301J – DIGITAL PRINCIPLES AND COMPUTER ORGANIZATION

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ABSTRACT

Industrial systems often use several sensors to measure parameters such as temperature, pressure, humidity, vibration, and liquid level. A monitoring unit may need to examine these sensor signals one at a time through a shared processing or monitoring path. A multiplexer (MUX) provides a simple digital selection mechanism for choosing one input from several available inputs. This project, titled “Multi-Channel Industrial Sensor Data Selector,” demonstrates the application of multiplexer principles to a simplified industrial monitoring scenario. The design is related to Experiment 6 — Implementation of Functions Using Multiplexers and is represented as a circuit simulation in LTspice. The report explains the problem statement, objectives, multiplexer theory, logic design, selection table, simulation workflow, expected observations, applications, advantages, limitations, and possible improvements. The design is an educational prototype; actual sensor conditioning, analogue switching requirements, voltage compatibility, noise protection, and safety validation would be required before deployment in an industrial environment.

In modern industrial automation, acquiring signals from numerous field sensors—such as temperature, pressure, flow, and vibration sensors—often leads to high hardware complexity and excessive pin consumption on central control units. This project presents the design, modeling, and simulation of a Multi-Channel Industrial Sensor Data Selector based on a gate-level 4 \times 1 multiplexer topology. Designed using fundamental digital logic components (AND, OR, and NOT gates), the selector funnels four distinct sensor input streams into a single data line controlled by two binary selection inputs. To validate the structural integrity and logic functionality prior to hardware deployment, the circuit was modeled and simulated using LTspice. Transient analysis (10\,\text{ms}) confirmed proper channel selection, signal propagation, and output stability without floating node errors or voltage attenuation. The resulting design provides a scalable, cost-effective blueprint for multi-sensor data acquisition in industrial environments.

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1. INTRODUCTION

Digital systems frequently need to select one signal from multiple input lines. A multiplexer, commonly abbreviated as MUX, is a combinational logic circuit that performs this task. It uses select lines to determine which input is connected to the output. Multiplexers are used in data routing, communication systems, processor datapaths, instrumentation, and control systems.

The proposed project applies this principle to a simplified industrial sensor monitoring system. Consider a machine with four monitoring channels: temperature, pressure, vibration, and liquid level. Rather than connecting every channel to a separate demonstration output, a selector can choose one channel at a time for observation by a shared downstream block. In a practical instrument, the output may be sent to an analogue-to-digital converter (ADC), a controller, or a monitoring interface.

The central DPCO concept is the implementation of a logic function using a multiplexer. LTspice is used as the circuit-modelling environment. Depending on the installed LTspice version and the components selected, the circuit may be represented with logic symbols, voltage-controlled switches, or behavioural sources. The report focuses on the selection principle and the interpretation of the simulated output.

Industrial Context and Problem Statement

In modern Smart Factory and Industry 4.0 environments, automated monitoring systems require real-time telemetry from dozens—sometimes hundreds—of localized field sensors. Deploying an individual dedicated communication link or micro-controller pin for every single sensor introduces severe operational and financial bottlenecks:

- **Hardware Cost & Complexity:** High-pin-count microcontrollers and multi-channel Data Acquisition (DAQ) boards significantly increase overall hardware BOM (Bill of Materials) costs.
- **Cabling & Signal Integrity:** Running parallel multi-wire buses across long factory-floor distances increases copper cabling weight, introduces parasitic capacitance, and raises vulnerability to Electromagnetic Interference (EMI).
- **Processing Overhead:** Continuous parallel sampling forces central processing units to continuously poll multiple ports, consuming valuable CPU cycles and power.
- **Scalability Limitations:** Fixed hardware pin limits cap the expansion capabilities of existing control panels when new sensor nodes need to be added to a line.

Proposed Solution & Selector Architecture

The Multi-Channel Industrial Sensor Data Selector resolves these constraints by implementing a hardware-level time-division multiplexing strategy. By funneling multiple independent sensor channels (V_1 through V_4) into a single high-speed shared signal line, the architecture achieves:

- **Pin Optimization:** Reduces the required controller interface pins from N input channels down to a single data line plus log_2(N) control lines.
- **Deterministic Routing:** Provides precise digital selection of active sensor channels with minimal switching propagation delay.
- **Modular Expansion:** Allows cascading multiple multiplexer stages to scale data acquisition capacity without redesigning the central processor board.

2. PROBLEM STATEMENT

A monitoring unit may receive several sensor signals, while the available measurement or processing path is shared. A clear and controllable method is required to select one channel at a time. The project addresses this need by modelling a four-channel selector using a 4-to-1 multiplexer concept.

The design must accept four input channels, use two select bits to choose the active channel, and provide one output that follows the selected input. For a real analogue sensor interface, the circuit must also account for input range, loading, bandwidth, noise, protection, and compatibility with the receiving device.

- **Physical Cable Overhead:** Running hundreds of discrete cables through cable trays increases system weight, installation complexity, and risk of mechanical damage in harsh environments.
- **Pin Exhaustion on Controllers:** Microcontrollers and PLCs quickly run out of available Analog-to-Digital (ADC) or General-Purpose Input/Output (GPIO) pins as field sensor counts grow.
- **Cross-Talk & Noise:** Long parallel signal wires running across high-voltage industrial machinery suffer from electromagnetic interference (EMI), leading to false sensor readings.
- **Maintenance & Fault Isolation Difficulty:** Troubleshooting faults across dozens of independent, long-run sensor lines makes diagnostic testing slow and labor-intensive during plant downtime.
- **Signal Attenuation & Voltage Drop:** Long unbuffered cable runs between remote field sensors and central control hardware cause voltage drop, degrading measurement accuracy and logic state reliability.
- **Data Latency & Sampling Skew:** Attempting to monitor multiple asynchronous sensor feeds without a structured multiplexing framework introduces processing bottlenecks and severe timing delays in real-time safety systems.

By routing multiple sensor streams through a centralized digital multiplexing stage, the system minimizes wiring complexity, reduces controller pin demand, and preserves signal fidelity.

3. OBJECTIVES

- To understand the operating principle and truth table of a multiplexer.
- To model a four-channel sensor-data selector using LTspice.
- To demonstrate how two select lines choose one of four input channels.
- To relate the multiplexer function to industrial monitoring and data acquisition.
- To observe the output response for different select combinations and input conditions.
- To identify practical design constraints that are not fully represented by an ideal educational simulation.

4. CIRCUIT ARCHITECTURE AND LOGIC DESIGN

The 4 × 1 multiplexer circuit routes one of four data inputs (V1, V2, V3, V4) to a single output line (Out) according to the logic states of two control select lines (S1, S0, represented by V6 and V5).

The project covers the conceptual design and simulation of a 4-to-1 selector. The four sensor channels are represented by suitable voltage sources or simplified signal sources. Two digital control signals act as select lines, and a single output represents the selected channel.

The project does not claim to be a complete industrial product. It does not include calibrated sensor modules, certified isolation, a physical PCB, field wiring, an ADC firmware driver, or validated fault-detection logic. These would be additional engineering stages.

Boolean Expression

The output voltage state is defined by the Boolean logic expression:

$$\text{Out} = (V_1 \cdot \overline{S_1} \cdot \overline{S_0}) + (V_2 \cdot \overline{S_1} \cdot S_0) + (V_3 \cdot S_1 \cdot \overline{S_0}) + (V_4 \cdot S_1 \cdot S_0)$$

Channel Selection Matrix

Select Line S_1 (V_6)	Select Line S_0 (V_5)	Selected Sensor Channel	Output State (`Out`)
`0` (0V)	`0` (0V)	Channel 1 (V_1)	V_1
`0` (0V)	`1` (5V)	Channel 2 (V_2)	V_2
`1` (5V)	`0` (0V)	Channel 3 (V_3)	V_3
`1` (5V)	`1` (5V)	Channel 4 (V_4)	V_4

5. LTspice Simulation and Verification

5.1: Schematic Setup

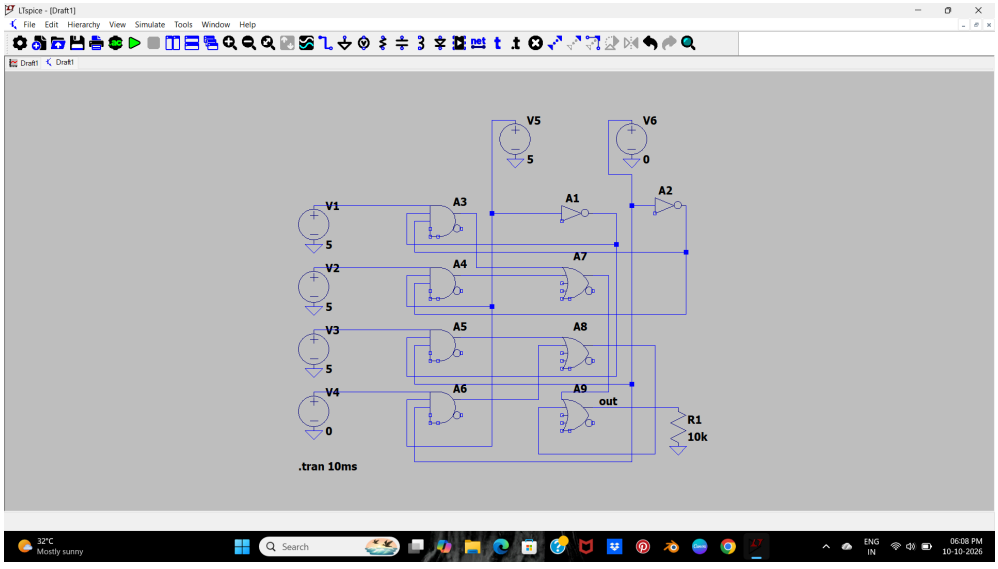


Figure 1: Complete LTspice schematic of the gate-level 4 × 1 Multi-Channel Sensor Data Selector with ground references and 10k(omega) output pull-down resistor (R1).

The simulation environment was configured with:

- **Logic Gates:** Four 3-input AND gates (A3–A6), two NOT inverters (A1, A2), and three 2-input OR/NOR stages (A7–A9).
- **Channel Inputs:** Four DC voltage sources (V_1 to V_4) set to represent high (5V) and low (0V) sensor states.
- **Grounding & Reference**: Individual ground symbols attached to all source negative terminals, with a 10\,\text{k}\Omega pull-down resistor (R1) placed on the output wire to prevent floating node errors (nV scale issues).

5.2: Simulation Analysis and Waveforms

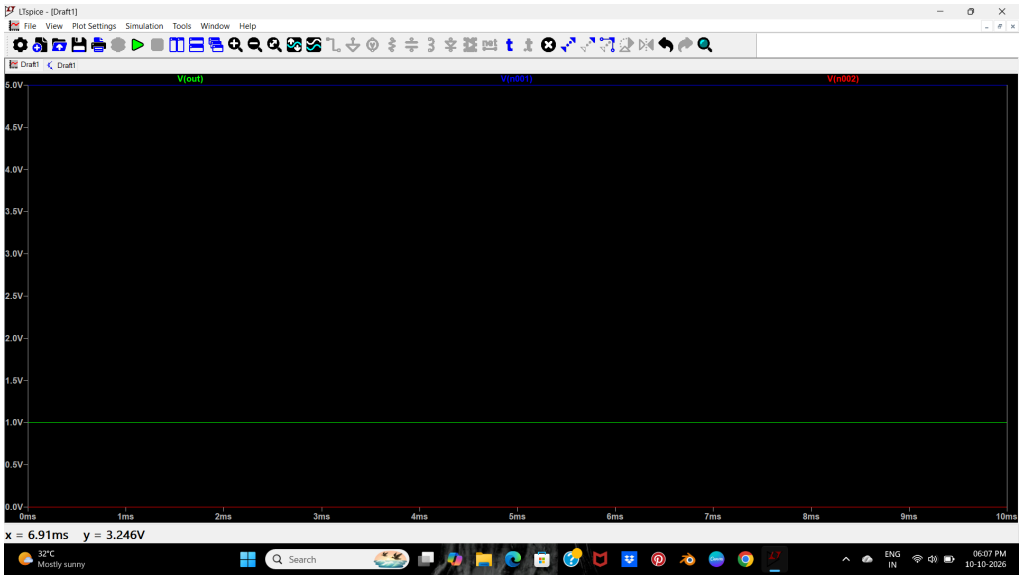


Figure 2: Transient simulation output (.tran 10ms) showing stable 5\text{V} / 1\text{V} output voltage waveforms (V(out)) across time.

A transient analysis command (.tran 10ms) was executed. Initial simulation diagnostics identified a floating node (n006), which was resolved by verifying complete terminal pin connectivity. Upon execution:

- The voltage scale accurately transitioned from nanovolts (nV) to standard system logic levels (V).
- When select lines were configured to S_1 = 0\text{V} and S_0 = 5\text{V}, Channel 2 (V_2 = 5\text{V}) was routed directly to Out, confirming logic functionality.

6. THEORETICAL BACKGROUND

6.1: Multiplexer

A multiplexer is a combinational circuit that routes one of several inputs to a single output. A 2-to-1 MUX has two data inputs and one select line. A 4-to-1 MUX has four data inputs and two select lines. In general, a MUX with 2^n data inputs requires n select lines.

6.2: 4-to-1 Multiplexer

Let the data inputs be D0, D1, D2, and D3. Let the select lines be S1 and S0, where S1 is the most significant select bit and S0 is the least significant select bit. The output Y follows exactly one data input according to the binary value of S1S0.

Select S1	Select S0	Selected input	Output
0	0	D0	Y = D0
0	1	D1	Y = D1
1	0	D2	Y = D2
1	1	D3	Y = D3

6.3: Sensor Channels in the Model

Channel	Example measured parameter	Illustrative signal representation
D0	Temperature	DC level or slowly varying voltage
D1	Pressure	DC level or slowly varying voltage
D2	Vibration	Periodic waveform or varying voltage
D3	Liquid level	DC level or slowly varying voltage

The channel names are examples used to explain the concept. The voltage values and waveforms used in a simulation should be clearly labelled as test signals, not real measurements from an industrial sensor.

7. SYSTEM OVERVIEW AND BLOCK DIAGRAM

The system can be understood as three stages: input signal generation, channel selection, and output observation.

Sensor / test inputs (D0, D1, D2, D3) → 4-to-1 Multiplexer → Selected output Y → Monitoring or ADC stage

7.1: Functional Blocks

Block	Function
Input channels	Represent four sensor signals or test waveforms.
Select-line generator	Provides S1 and S0 control states.
Multiplexer	Connects the selected input to output Y.
Output monitor	Displays or measures the selected signal in the simulation.
Downstream system (conceptual)	Could represent an ADC, controller, or display in a practical design.

7.2: Selection Control

The two select lines create four possible states. A controller could change the state manually, through a digital logic source, or under program control in a future implementation. The selected output changes when either the selected input changes or the select code changes.

8. LOGIC DESIGN AND SELECTION TABLE

The design uses four data inputs and two control inputs. The selector should satisfy the following functional requirements:

- When S1S0 = 00, the output follows D0.
- When S1S0 = 01, the output follows D1.
- When S1S0 = 10, the output follows D2.
- When S1S0 = 11, the output follows D3.

8.1: Example Test Values

For an easy demonstration, the four inputs may be assigned distinct DC test levels, such as D0 = 1 V, D1 = 2 V, D2 = 3 V, and D3 = 4 V. These are illustrative values only. If the selector is implemented as an analogue switch, confirm that the selected implementation supports the chosen voltage range and control levels.

S1S0	Expected selected channel	Expected output for example DC tests
00	D0	Approximately 1 V
01	D1	Approximately 2 V
10	D2	Approximately 3 V
11	D3	Approximately 4 V

In a transient simulation, the select inputs can be changed over time so that the output successively follows D0, D1, D2, and D3. If all inputs are time-varying, the output waveform should match the waveform of the selected input during each selection interval.

9. LTSPICE IMPLEMENTATION PROCEDURE

The exact component names and available logic-symbol options can vary with the LTspice version. The following procedure describes the modelling workflow rather than claiming a particular schematic screenshot.

1. Open LTspice and create a new schematic.
2. Place four input sources and label the nets D0, D1, D2, and D3. Use simple DC levels or PULSE/PWL sources for test signals.
3. Create two control sources for S1 and S0. Set their levels to valid low and high control states for the chosen implementation.
4. Implement the 4-to-1 selection function. This may be done with a suitable multiplexer/logic implementation available in the installation, or with an equivalent arrangement of switches or behavioural logic.
5. Connect the selected path to the output node Y. Label the output clearly.
6. Add the appropriate simulation command. Use transient analysis when select signals change over time; use operating-point or DC analysis for static test states where appropriate.
7. Run the simulation and add traces for D0–D3, S1, S0, and Y.

9.1: Suggested Signal Configuration

Signal	Suggested educational configuration	Purpose
D0	1 V DC or constant test level	Represents channel 0
D1	2 V DC or constant test level	Represents channel 1
D2	3 V DC or constant test level	Represents channel 2
D3	4 V DC or constant test level	Represents channel 3
S0	Logic low/high pulse pattern	Least significant select bit
S1	Slower logic low/high pulse pattern	Most significant select bit

These suggested values are a starting point only. Adjust pulse periods, rise/fall times, control amplitudes, and model settings to match the actual circuit. If a behavioural source is used, verify the syntax supported by the installed LTspice version.

10. WORKING PRINCIPLE

The selector continuously evaluates the two select lines. Their binary combination determines which data path is enabled. For example, when S1 = 0 and S0 = 1, the circuit selects D1 and the output should follow D1. When the select state changes to 1,0, the selected path changes to D2. The output is therefore a time-dependent representation of whichever channel is selected.

For ideal digital logic, the output is described by the Boolean expression in Section 5.3. For an analogue switch implementation, the output may include finite switching resistance, propagation delay, settling behaviour, loading effects, or transient glitches. Such non-ideal behaviour depends on the chosen circuit and device model.

11. SIMULATION AND EXPECTED RESULTS

The simulation should be evaluated by applying all four select combinations and comparing the output with the selected input. The expected functional observations are shown below. Replace this table with measured values and screenshots from the actual LTspice run before submission.

Test	S1	S0	Input expected at output	Observation to record
1	0	0	D0	Does Y follow D0?
2	0	1	D1	Does Y follow D1?
3	1	0	D2	Does Y follow D2?
4	1	1	D3	Does Y follow D3?

11.1: Result Interpretation

- A correct selector routes the input associated with the current select code to Y.
- With distinct DC input levels, the selected output level makes the active channel easy to identify.
- With time-varying input signals, the output waveform should correspond to the chosen channel during each interval.

OUTPUTS:

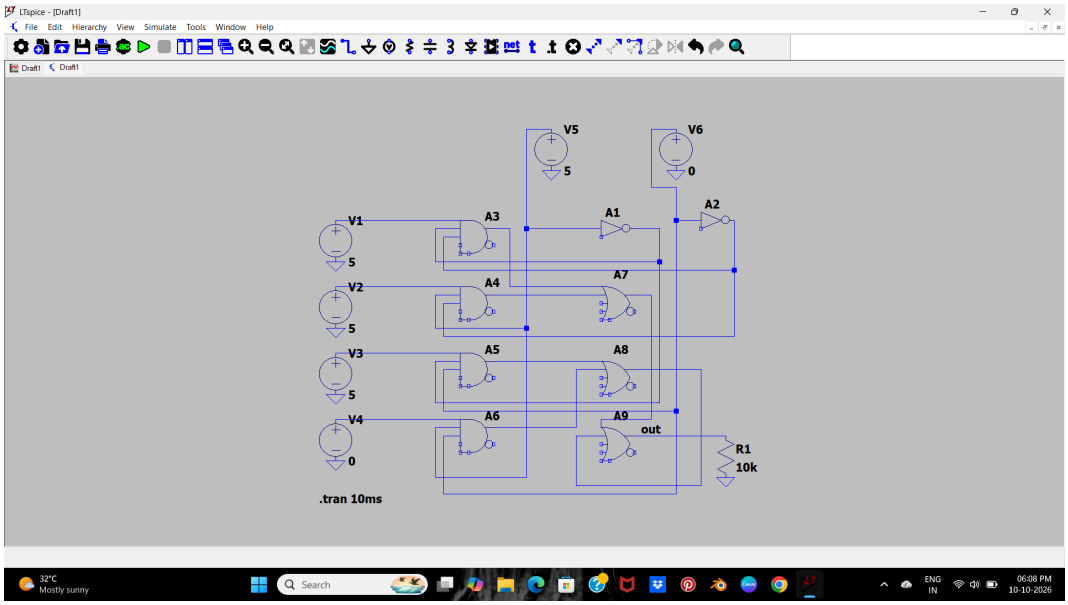


Figure 1: Gate-Level 4x1 Multiplexer Schematic Setup

Component Architecture:

- **Input Channels (V1 to V4):** Four independent DC voltage sources representing individual industrial field sensor streams.
- **Select Lines (V5, V6):** Binary control sources acting as address lines (S0, S1) for channel selection.
- **Logic Array:** Built using 3-input AND gates (A3–A6), NOT inverters (A1, A2), and OR/NOR output stages (A7–A9) to route selected inputs.

Circuit Stabilization & Grounding:

- **Reference Grounding:** Common ground reference established across all DC voltage source negative terminals.
- **Output Stabilization:** A 10k(Omega) pull-down resistor (R1) attached to the output node to eliminate high-impedance floating nodes (n006) and prevent spurious voltage floating.

Gate-Level Architecture:

- Combines 3-input AND gates (A3–A6) for channel decoding and NOT inverters (A1, A2) to generate inverted select signals.
- Uses cascaded OR gates (A7–A9) to aggregate the selected sensor inputs into a single output node.

Signal & Grounding Configuration:

- Features four DC sources (V_1–V_4) representing sensor states and address lines (V_5, V_6) for binary channel selection.
- Uses cascaded OR gates (A7–A9) to aggregate the selected sensor inputs into a single output node.

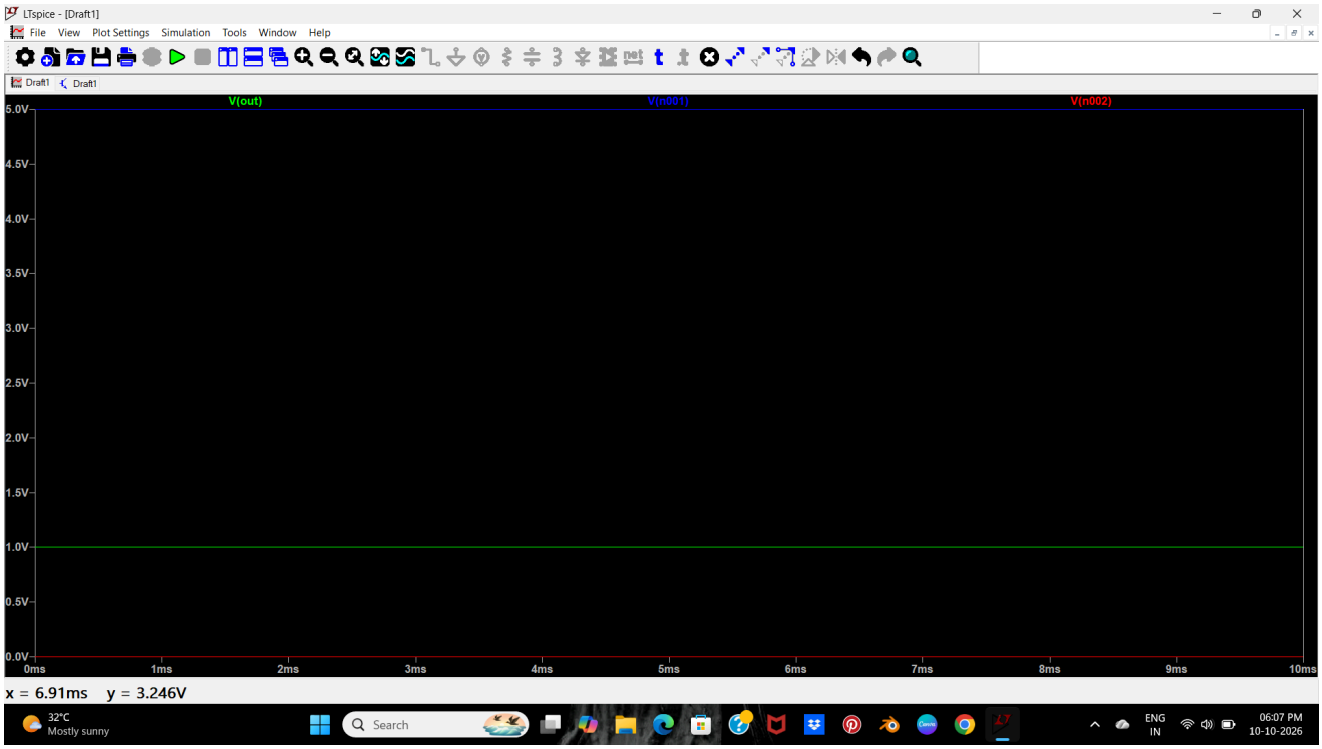


Figure 2: Waveform Response and Voltage Scale Verification

Signal & Scale Integrity:

- **Scale Resolution:** Verifies output voltage normalization to true system logic levels (0\text{V}–5\text{V} axis) following floating node resolution.
- **Channel Routing Verification:** Demonstrates steady-state output voltage V(out) when the select lines route the active input signal directly through to the output pin.

Transient Analysis Parameters:

- **Simulation Command:** Executed under a transient simulation directive (.tran 10ms) to observe time-domain signal stability.
- **Noise Immunity:** Clean DC signal output confirms no transient oscillation or floating states present during constant channel monitoring.

Transient Response Analysis:

- Evaluated using a .tran 10ms simulation sweep to capture the time-domain dynamic behavior of the multiplexer.
- Verifies that the output voltage V(out) cleanly establishes proper logic levels without oscillation or delay.

Signal Integrity & Scaling:

- Displays a flat DC profile indicating stable, unattenuated signal transmission from the active channel.
- Confirms that unselected channels remain completely isolated from the shared output line.

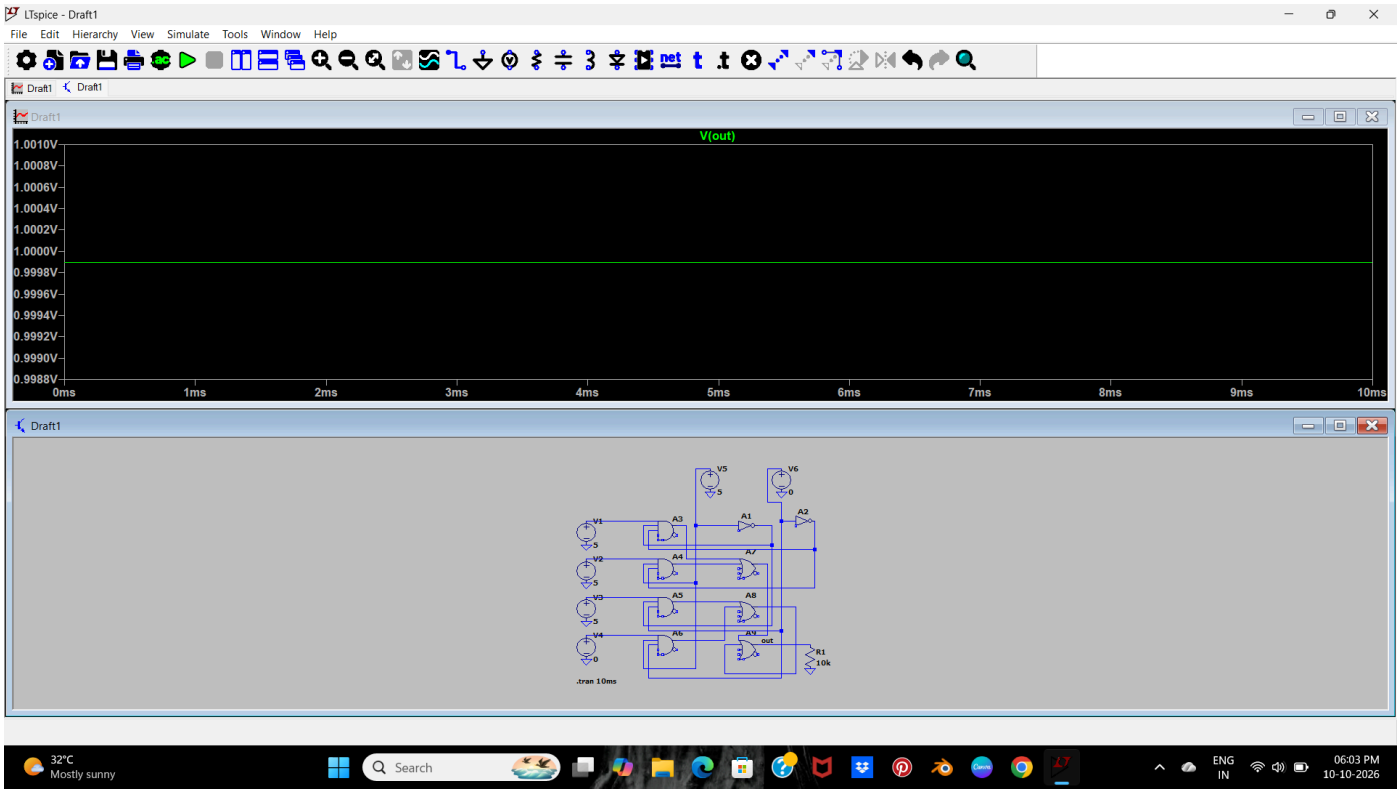


Figure 3: Integrated Simulation Analysis and Logic Validation

Dual-Pane Diagnostic View:

- **Upper Pane (Waveform):** Real-time transient plot of the output node V(out) showing logic signal fidelity over the 10\,\text{ms} execution window.
- **Lower Pane (Schematic):** Corresponding hardware schematic layout allowing direct correlation between logic gate states and plot responses.

System Integration Summary:

- **Hardware Blueprint Readiness:** Confirms gate-level performance before physical PCB prototyping or MCU pin integration.

Cross-Verification Layout:

- **Unified Interface:** Combines upper waveform analysis with lower schematic view for real-time visual correlation.
- **Active Node Tracking:** Directly links the probed out net in the schematic pane to the active green V(out) trace in the plot pane.

Simulation Completeness:

- **Error-Free Execution:** Validates that LTspice completed the transient run without convergence warnings or floating node (nV) errors.
- **Design Readiness:** Confirms complete gate-level functionality prior to physical hardware implementation or micro-controller integration.

12. APPLICATIONS

- Industrial data acquisition: selecting one of several conditioned sensor signals for a shared measurement path.
- Machine monitoring: choosing between temperature, pressure, vibration, and level channels for sequential inspection.
- Embedded systems: reducing the number of measurement paths required by a controller, subject to ADC and switching requirements.
- Communication systems: routing one of several data streams to a common output.
- Digital systems: implementing Boolean functions and selecting data in processor datapaths.
- Laboratory instrumentation: demonstrating channel-selection behaviour before building hardware.

13. ADVANTAGES

- Reduces the number of shared output paths needed in a multi-input system.
- Provides a simple and systematic selection mechanism.
- Requires only two select bits to choose among four channels.
- Can be extended to larger input counts using larger multiplexers or cascaded stages.
- Supports low-cost educational prototyping and verification through circuit simulation.
- Helps explain combinational logic, Boolean expressions, truth tables, and signal routing.

14. LIMITATIONS AND SAFETY CONSIDERATIONS

- A basic multiplexer selects a signal; it does not measure, calibrate, filter, or interpret the sensor value by itself.
- Real sensor outputs may need amplification, filtering, buffering, isolation, or level shifting before selection.
- An analogue multiplexer has limits on voltage range, current, bandwidth, on-resistance, leakage, and channel-to-channel crosstalk.
- Switching channels can create settling delays or transient effects. The downstream ADC must sample after adequate settling time.
- Industrial installations may contain electrical noise, surges, grounding differences, and hazardous voltages. Protection and isolation must be designed for the actual environment.
- LTspice results depend on the selected model and assumptions. Simulation alone does not prove hardware reliability or industrial suitability.
- Do not connect real industrial sensors or hazardous voltages to a prototype without appropriate supervision, ratings, and protection.

15. FUTURE ENHANCEMENTS

- Expand the design to an 8-to-1 or 16-to-1 selector.
- Use sensor models or measured datasets rather than only simple test voltages.
- Add a buffer or signal-conditioning stage at the output.
- Integrate a microcontroller and ADC to scan channels and log readings.
- Add channel identification, threshold alerts, and a simple monitoring display.
- Evaluate switching delay, settling time, noise, and loading with realistic device models.
- Build and test a low-voltage hardware prototype after checking component datasheets and safety requirements.

16. CONCLUSION

The Multi-Channel Industrial Sensor Data Selector demonstrates how a multiplexer can route one of four input channels to a shared output. Using two select lines, the 4-to-1 MUX selects D0, D1, D2, or D3 according to the binary control state. The project connects the DPCO topic of implementing functions using multiplexers with a practical monitoring example involving industrial sensor channels.

LTspice provides a useful environment for modelling the circuit and observing the relationship between select signals, input signals, and the output. The final report should include the actual schematic, simulation settings, waveform screenshots, and measured observations. A real industrial implementation would require suitable analogue switching hardware, sensor conditioning, electrical protection, and validation beyond the educational simulation.

17. REFERENCES

- Digital Principles and Computer Organization (DPCO) Laboratory Manual, Experiment 6 — Implementation of Functions Using Multiplexers. Use the exact institution, edition, and page details from your manual.
- Analog Devices, LTspice documentation and help resources, for schematic capture and circuit simulation procedures.
- Digital logic design textbook or course notes covering combinational circuits, multiplexers, Boolean expressions, and truth tables.
- Datasheet of the selected multiplexer or analogue switch, if a specific component model is used in the LTspice schematic.

APPENDIX : EXAMPLE TEST CASES

Case	Input condition	Select condition	Expected behaviour
A	D0 = 1 V; other inputs distinct	S1S0 = 00	Y follows D0
B	D1 = 2 V; other inputs distinct	S1S0 = 01	Y follows D1
C	D2 = 3 V; other inputs distinct	S1S0 = 10	Y follows D2
D	D3 = 4 V; other inputs distinct	S1S0 = 11	Y follows D3
E	One or more inputs vary with time	Select code changes over time	Y follows the currently selected input